Optical matching and beam loss in a booster-to-storage-ring transfer line for nonlinear-kicker injection at the Korea Light Source

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Abstract

We study optical matching and beam loss in a design version of the Korea Light Source booster-to-storage-ring transfer line for nonlinear-kicker injection. The line has three bending magnets and nine quadrupoles. Matching fixes the nominal beam sizes at both ends, so lowering the interior peak beta leaves the nominal clearance at the limiting injection-septum blade unchanged. After static correction through existing dipole field trims, booster shot jitter dominates at the assumed stress level. In the nominal matched model, horizontal phase advance determines its partition between exit position and angle. A Gaussian loss model checked against particle tracking places the smallest blade loss at the lower edge of the matched phase range. On 200 independent steered error realizations, the best-found setting has a mean jitter-integrated loss 4.48 times smaller than the matched reference (95 % interval 4.33–4.63), at larger exit-angle jitter. This linearized Gaussian-model prediction assumes position and angle jitter of 1.3 and 2.2 beam sizes; its interval reflects only sampling of the error realizations. At 0.1–0.2 of that level, blade loss is negligible for every matched setting. Steering then leaves about 0.18 mrad rms in horizontal angle and 1.07 mm in vertical position; the model has no vertical corrector. A dispersion-free exit lowers blade loss further. Storage-ring capture is outside the scope of this study.

Keywords: Beam dynamics, Transfer line, Optical matching, Injection jitter, Beam loss, Korea Light Source

1. Introduction

Diffraction-limited storage rings require control of the beam distribution delivered to the injection region [1]. The booster-to-storage-ring (BTS) transfer line connects the extracted booster beam to the prescribed injection conditions of the storage ring. Its optical design must accommodate the incoming emittance and momentum spread, satisfy the exit lattice functions, and maintain adequate clearance through the magnets and injection septum. At the septum, the beam envelope and trajectory must fit within a channel bounded by the blade; downstream, both position and angle affect the conditions delivered to the injection kicker. The 4 GeV Korea Light Source (KLS), described in its technical design report (TDR) [2] and in Refs. [3, 4], provides the application considered here. In the TDR design, the BTS line consists of two dipole and eight quadrupole magnets, and the storage ring is filled by off-axis injection with a four-kicker orbit bump and an in-vacuum thin septum with a 1 mm blade [2]. The line studied here is a different design version of the BTS, with three main dipoles and nine quadrupoles (Section 2), and its injection geometry is taken from our study of nonlinear-kicker magnet (NKM) injection into the KLS storage ring [5]; the results apply to this version of the line.

Previous transfer line studies establish the importance of treating the accelerator interfaces and available geometry together. The HEPS high-energy transport-line design matches the

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injection-point lattice functions while accommodating the relative positions of the booster and storage ring, equipment clearances, and the selected injection scheme [6]. A subsequent HEPS study updates the transfer lines in response to the evolving accelerator layouts and optics [7]. For Diamond-II, the transfer lines are designed to preserve bunch properties with low loss while retaining diagnostic capability and flexibility for matching; the BTS modifications connect the new booster extraction geometry to the storage-ring septum [8]. The ESRF–EBS rematching similarly includes injection-region fields and mechanical constraints [9]. These examples show why an optical target and the physical restrictions along the transport path must be specified together.

Beam characterization and trajectory control address different parts of this matching problem. The SLS 2.0 transfer line includes a nondispersive section for quadrupole-scan measurements and a corrector configuration for separate control of injection position and angle [10]. Its commissioning results connect beam characterization to downstream matching and trajectory stabilization. The Sirius injection study reports efficiencies separately for the injector stages and examines booster realignment, emittance exchange, and remaining reproducibility issues [11]. Separating these stages prevents a transfer-line improvement from being equated with the same change in storage-ring injection efficiency. It also motivates evaluating a nominally matched line under incoming beam fluctuations and residual trajectory errors, rather than relying on envelope plots alone.

The requirements imposed by the injection device provide an additional design constraint. Pulsed-sextupole injection at MAX IV relates the incoming beam distribution to the nonlinear injection process [12]. A recent TPS study proposes a new booster-to-storage-ring line for nonlinear-kicker injection and identifies existing-ring geometry, trajectory correction, and injected-beam characterization as design considerations [13]. Together, these studies motivate distinguishing nominal optical matching, clearance at the limiting restriction, and sensitivity to the position and angle delivered to the injection system. The present study examines those distinctions within a fixed BTS model with explicit aperture and jitter models.

A common transfer-line design objective is to reduce the peak beta functions and thereby increase aperture margin. For fixed emittances, momentum spread, and matched entrance and exit lattice functions, however, quadrupole rematching preserves the nominal beam sizes at the boundaries. Restrictions in the fixed sections upstream and downstream of the quadrupoles can therefore limit the line even when its interior envelopes are reduced. The same boundary conditions leave freedom in the betatron phase advance, which changes the transport of booster position and angle jitter. A matched setting can consequently change the loss probability at a septum blade without changing its nominal clearance. Distinguishing these mechanisms is necessary to determine whether a proposed optics improvement addresses the limiting restriction or only redistributes margin elsewhere in the line.

The goal of this study is to identify which properties of a matched BTS setting control the beam loss at the active restriction, and what a designer can choose among matched settings. The baseline setting is the nine-quadrupole configuration supplied with the line before matching; it does not satisfy the prescribed exit and serves only as the starting point. Bounded optical matching followed by vertical-beta optimization gives the optimized low-beta setting (low beta). A matched intermediate setting, referred to below as the matched reference, is the comparison for all error studies. The line is modeled with element-wise apertures and the injection-septum blade geometry of the NKM injection study [5]. The subsequent error study compares the matched reference, low-beta, and loss-based settings after static trajectory correction using field trims of existing dipoles. A Gaussian loss probability derived from the transported beam and centroid covariance is validated against particle tracking and used to select a loss-based setting before evaluation on independently seeded error realizations. The analysis also examines the phase advance of the matched settings, a dispersion-free exit target, and the dependence on the error levels, in particular on the booster jitter. The study ends at the transfer-line exit; storage-ring

capture and NKM transparency require the subsequent injection calculation.

Section 2 defines the model, Section 3 the criteria and their validation, Section 4 the results, and Section 5 the design implications and limitations.

2. Parametric model and physical observables

2.1. Lattice, coordinates, and boundary conditions

The baseline BTS model contains 36 elements over 21.789 m, from marker m1 at the entrance to marker m4 downstream of the injection septum. Beam dynamics simulations use the Accelerator Toolbox (AT) [14] through its Python implementation pyAT [15]. The line has nine 0.2 m quadrupoles in three triplets, three main dipole bending magnets (`b1`, `b2`, and `b3`, in beam order), the booster extraction kicker `kext`, the extraction septum `sept_in` and the injection septum `sept_ex`; the kicker and both septa are modeled as dipole elements (Fig. 1). Quadrupoles are labeled `qij`, where i identifies the triplet and j the magnet within that triplet, both in beam order. Thus `q11` is the first quadrupole in the first triplet and `q33` is the last quadrupole in the third triplet. Local coordinates are $(x, p_x, y, p_y, \delta, ct)$ with positions/path length in m, normalized canonical momenta approximately angles in rad, and $\delta = \Delta p/p_0$ dimensionless. The six design bend angles are 7.5 mrad, 88.5 mrad, −111.701 mrad, 176 mrad, −111.701 mrad and 88.5 mrad; reference geometry is held fixed when fields or particle momenta change.

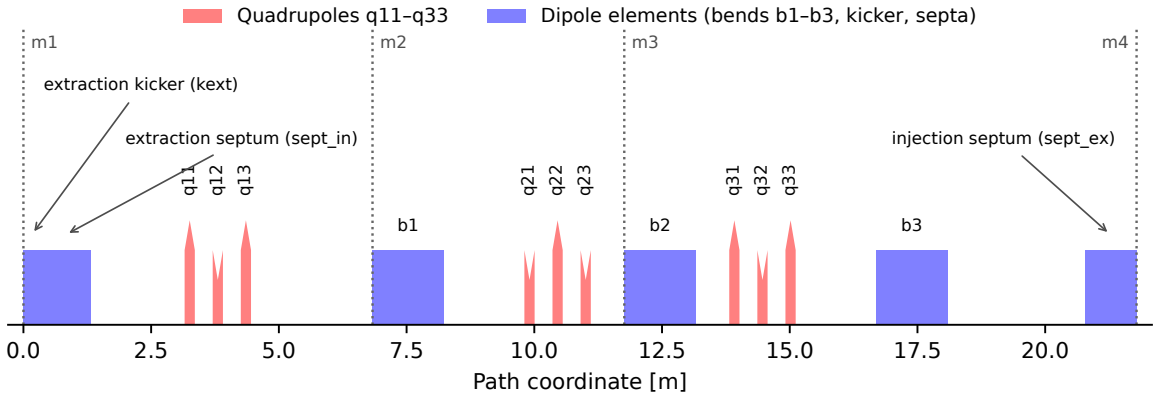


Figure 1: Baseline BTS element sequence drawn with Accelerator Toolbox synoptic plotting. Red symbols are the quadrupoles `q11`–`q33`; blue blocks are the dipole elements: the extraction kicker `kext`, the extraction septum `sept_in`, the main bends `b1`–`b3` and the injection septum `sept_ex`. Blank intervals are drifts. Dotted lines mark the markers m1–m4 used for boundary and intermediate optics observations.

Table 1: Prescribed entrance and exit lattice functions. The exit values describe the injected beam at the injection-septum exit used in the NKM injection study [5]. A dispersion-free exit is examined in Section 4.4.

Observable	m1 entrance	m4 target
β_x, β_y [m]	7.560000, 12.269000	2.336495, 4.256241
α_x, α_y [1]	1.523100, −1.654700	−0.016335, 0.017772
D_x [m], D'_x [1]	0.276200, −0.065700	0.080868, 0.047472

Entrance and exit lattice functions (Table 1) are inherited design values. The exit m4 is the injection-septum exit: the NKM injection study uses exactly these values as its injected beam and transports it 2.26 m to the kicker. The configured nonzero exit dispersion (80.9 mm, 47.5 mrad) differs from the ring dispersion at the injection kicker, which is about 0.3 mm; we therefore also examine a dispersion-free exit. Quadrupole family bounds, bore radius, pole-field

limit, the 60 m beta screening ceiling and square aperture limits are model parameters. The BTS line is independently validated using analytical maps of the same AT element sequence.

An earlier four-station model, with square apertures only at markers m1–m4 and none on the magnets, is kept in the supplement for comparison.

2.2. Beam covariance and clearance

This study uses the booster horizontal design input $\epsilon_x = 7.67\,\mathrm{nm\,rad}$ and $\sigma_\delta = 0.106\,\%$ [4]. The vertical input $\epsilon_y = 0.767\,\mathrm{nm\,rad}$ corresponds to a vertical-to-horizontal emittance ratio of 10 %. Betatron coordinates and momentum are independent untruncated Gaussians before dispersion is added; initial transverse coupling is zero and $ct = 0$ for this RF-free single pass. The transverse blocks of the complete covariance are

$$\Sigma_{u,\mathrm{bet}} = \epsilon_u \begin{pmatrix} \beta_u & -\alpha_u \\ -\alpha_u & (1+\alpha_u^2)/\beta_u \end{pmatrix}, \qquad \Sigma_0 = \Sigma_{\mathrm{bet}} + \sigma_\delta^2 \mathbf{d}\mathbf{d}^\mathsf{T}, \tag{1}$$

where $u \in \{x, y\}$ denotes the transverse plane, β_u and α_u are the Courant–Snyder parameters and ϵ_u the emittance in that plane, Σ_{bet} is the 6×6 betatron covariance with the two 2×2 blocks $\Sigma_{u,\mathrm{bet}}$ on its diagonal, and $\mathbf{d} = (D_x, D'_x, D_y, D'_y, 1, 0)^\mathsf{T}$. The same index u is used for the plane in all equations below. Rolls of the quadrupoles about the beam axis (Section 2.4) couple the horizontal and vertical planes downstream of each rolled magnet. Full covariance transport, $\Sigma_s = M_s \Sigma_0 M_s^\mathsf{T}$, retains that coupling and momentum correlation. A local affine map gives the centroid $\bar{\mathbf{z}}_s = M_s \bar{\mathbf{z}}_0 + \mathbf{b}_s$.

For each aperture station s and plane u, with lower and upper aperture edges a_- and a_+, we define

$$c_u(s) = \min[a_+ - \bar{u}(s), \bar{u}(s) - a_-] - 5\sqrt{\Sigma_{uu}(s)}, \qquad c_{\min} = \min_{s \in \mathrm{stations}, u \in \{x,y\}} c_u(s). \tag{2}$$

Here $\bar{u}(s)$ is the beam centroid and $\sqrt{\Sigma_{uu}(s)}$ the projected rms beam size, including the dispersive contribution. The first term in Eq. (2) is the distance from the centroid to the nearer aperture edge; subtracting five rms beam sizes gives the margin $c_u(s)$ that remains between the five-sigma beam envelope and that edge. A positive c_u means that the envelope fits inside the aperture, $c_u = 0$ that it touches the edge, and a negative c_u that it overlaps the edge by $|c_u|$. The minimum $c_{\min}$ over all stations and both planes identifies the limiting restriction. The five-sigma envelope is a conventional coverage measure of the projected distribution; an untruncated Gaussian beam is never completely contained. Because the beam size scales as $\sqrt{\beta_u}$, a reduction of the beta function by some metres does not increase the clearance by a comparable length: the clearance changes only by five times the change of the rms size at the limiting station, and not at all where the optics are fixed by the boundary conditions (Section 3).

The quadrupole strengths are bounded so that every setting considered remains within the field range of conventional quadrupole magnets. At 4 GeV the magnetic rigidity is $B\rho = p_0/e \approx 13.342\,56\,\mathrm{T\,m}$. A normalized strength K in m^{-2} corresponds to the field gradient $G = K(B\rho)$ in $\mathrm{T\,m}^{-1}$, and a quadrupole with bore radius r then has the pole-tip field $B_{\mathrm{pole}} = |G|r$ in T. The strength bounds are $|K| \le 2.5\,\mathrm{m}^{-2}$ for the outer quadrupoles q11 and q33 and $|K| \le 3\,\mathrm{m}^{-2}$ for the other seven. With $r = 19.35\,\mathrm{mm}$, the largest gradients are $33.356\,\mathrm{T\,m}^{-1}$ and $40.028\,\mathrm{T\,m}^{-1}$, and the corresponding pole-tip fields are 0.645 T and 0.775 T, respectively. Both remain below the 1.2 T pole-tip limit used for screening. This check confirms that the optimized strengths are consistent with the magnet model; it is not an engineering design of the magnets.

2.3. Element-wise restrictions

Every element of the BTS line carries its own aperture (Table 2), and AT checks each aperture at the element entrance; the line end (m4) carries the injection-septum limits, so the blade is checked at both ends of the septum. Particle losses are therefore counted at these declared

check points, not as continuous interception along a chamber or channel. Interior envelopes are sampled for the nominal profile on an observation-only copy of the line, with four subdivisions per quadrupole and eight per bend or septum; these nominal samples locate the interior minimum clearance but do not certify the beam tails under errors.

The injection-septum blade separates the main-field channel traversed by the injected beam from the stored-beam region, where a small magnetic field is required. Its geometry follows the NKM injection study [5]; for comparison, the baseline KLS injection uses an in-vacuum thin septum with a 1 mm blade [2]. Let X denote the horizontal coordinate relative to the storage-ring reference orbit at the septum exit. The blade face toward the stored beam is at $X = -16$ mm and its thickness is 2 mm, so the face toward the injected beam is at $X_b = -18$ mm. The nominal injected-beam centroid is at $X_{\mathrm{inj}} = -19.478$ mm, giving the centroid-to-blade distance

$$d_b = X_b - X_{\mathrm{inj}} = 1.478\,\mathrm{mm}. \tag{3}$$

For the configured beam, this distance is approximately three rms horizontal beam sizes plus 1 mm. In the local BTS coordinate at the exit, $x = X - X_{\mathrm{inj}}$, the nominal injected beam is centered at $x = 0$ and the blade face is at $x = +d_b$. This boundary is applied on the side facing the stored beam in the injection septum and at the line end (marker m4). The five-sigma clearance in Eq. 2 additionally subtracts the projected beam envelope and includes any centroid displacement; it is distinct from d_b.

The injected beam is intentionally deflected by the septum's main magnetic field while it travels through the injection channel. Proximity to the blade does not imply that the beam is in a field-free region. In the BTS model, the septum's bending field is represented by a dipole element, while the blade face is represented separately by the aperture boundary. Detailed spatial fringe-field and leakage-field distributions are not resolved. The blade dimensions and offset are study parameters rather than established hardware specifications; the other aperture dimensions are model parameters, and the bend gap and the extraction-septum aperture are swept (Section 4).

Bend chambers are taken as curved: the sagitta of the reference arc over its chord is 30.8 mm in b2 and 19.5 mm in b1/b3, so a straight chamber would need that much extra width. Clearances (Eq. 2) are evaluated at every aperture, with the projected halfwidth of a round aperture reduced by the other plane's centroid.

Table 2: Element-wise aperture model, reported as halfwidths. For the injection septum, a_x is the nominal injected-centroid-to-blade distance on the $+x$ side, not the blade thickness or its offset from the storage-ring reference orbit. The outer wall is at -8.5 mm in local BTS coordinates.

Region	Shape	a_x [mm]	a_y [mm]
Drifts	round	19.350	19.35
Quadrupoles q11–q33	round	19.350	19.35
Extraction kicker	rect.	15.000	10.00
Extraction septum	rect.	7.500	5.00
Bends b1–b3 (curved)	rect.	30.000	10.00
Injection septum	one-sided	1.478	5.00

2.4. Errors, jitter and trajectory correction

Static machine errors are independent Gaussian errors of the nine quadrupoles, namely gradient errors (0.1 % rms), x/y offsets (100 µm rms) and rolls about the beam axis (0.5 mrad rms), together with field errors of the six dipole elements (0.05 % rms). Each independent draw of these static errors defines one error realization of the line. Each shot adds booster jitter at m1: entrance beta factors (5 %), centroid position 0.5 mm, angle 0.2 mrad and momentum 0.1 %, the latter carried by the entrance dispersion. These levels are model inputs; the booster jitter

is the same as in the NKM injection study [5]. In beam-size units the betatron jitter is large: at m1 the rms beam is 0.379 mm and 91 µrad, so the position and angle jitter are 1.3 and 2.2 beam sizes, and the mean centroid action is 54 times the emittance. No KLS specification of the booster extraction stability is available to us, so we treat these values as a stress level and report how the results depend on them. The sensitivity to all error levels is examined in Section 4.5.

Static trajectory errors are corrected without changing the line: no corrector or monitor element is added. The horizontal knobs are field trims of the six existing dipoles, applied as additional fixed-geometry dipole fields like the dipole errors. Horizontal readings are taken at existing element boundaries, after each quadrupole, at the injection-septum entrance and at the line end, with a static 50 µm rms reading error (offset and calibration residual). The lattice model contains no monitor elements, so these eleven readings are assumed instrumentation, and the six trims are assumed to be independently powered; fewer readings and series-powered bends are examined in Section 4.2. The trims minimize the weighted readings (septum entrance and end weighted 3) with the response of the error-free model line. With $\mathbf{r}$ the vector of readings, A the response matrix of the readings to the trims $\boldsymbol{\theta}$, and W the diagonal weight matrix, the trims minimize

$$\|W(\mathbf{r} + A\boldsymbol{\theta})\|^2 + \lambda^2\|\boldsymbol{\theta}\|^2. \tag{4}$$

The second term is a Tikhonov (ridge) regularization [16]: it penalizes large trims and keeps the solution stable when the response matrix is ill-conditioned. Here λ is 10^{-3} times the largest singular value of WA. The trims are limited to ± 1 mrad of kick, and the correction is applied in two passes. The correction uses no knowledge of the actual errors and does not act on shot jitter. The line has no vertical knob, so the vertical plane is not corrected. Every error realization is evaluated uncorrected and corrected on identical particles. Each monitor location has one static reading error per error realization, and every setting and steering layout of that realization uses it, so paired comparisons share the magnet errors, readings, shot jitter and particles. The earlier steering study (supplement Section 5) used reading errors drawn separately for each setting.

3. Physical criterion, validation and comparison protocol

3.1. Active restrictions and boundary-fixed clearances

The two end stations sit where the optics are fixed by boundary data: the entrance Courant–Snyder parameters and dispersion are inputs, and every admissible setting matches the exit to its target. For a centered nominal beam the clearance there is therefore $c_u = a - 5\sqrt{\epsilon_u\beta_u + D_u^2\sigma_\delta^2}$ with the boundary values of Table 1, for every quadrupole setting K that satisfies the match. With $a = 19.35$ mm this gives 17.455 mm/18.865 mm at the line entrance (m1) and 18.555 mm/19.064 mm at the exit (x/y), the values found numerically in the four-station model, where these end stations limit every matched setting. This statement concerns the nominal, error-free, exactly matched line. Under errors the exit covariance $\Sigma_{\text{out}} = M(K,e)\Sigma_{\text{in}}(e)M(K,e)^{\mathsf{T}}$ also depends on K, through gradient errors, rolls, chromatic transport of off-momentum particles and the sampled entrance beta factors; together with the centroid response, these make the exit clearance K-dependent. The same nominal statement holds for any restriction downstream of q33, whereas restrictions upstream of q11 are independent of K even with errors. All error-study results below transport the full perturbed covariance and centroid of each error realization.

At a one-sided edge a_+, such as the septum blade, $c_u = a_+ - \bar{u} - 5\sigma_u$ with $\sigma_u = \sqrt{\Sigma_{uu}}$, and the local sensitivity to a quadrupole strength is

$$\frac{\partial c_u}{\partial K_j} = -\frac{\partial \bar{u}}{\partial K_j} - \frac{5}{2\sigma_u}\frac{\partial \Sigma_{uu}}{\partial K_j}. \tag{5}$$

At a symmetric aperture the first term carries the sign of $\bar{u}$; switching restrictions or crossing zero centroid makes the expression piecewise. In an uncoupled plane $\Sigma_{uu} = \epsilon_u\beta_u + D_u^2\sigma_\delta^2$; our coupled

calculation retains the complete covariance. Reducing beta controls one size contribution, not necessarily dispersion or orbit sensitivity. We export these variances and centroids separately, adding variances rather than RMS sizes.

3.2. Jitter transfer between matched ends

For the error-free, uncoupled line with both ends exactly matched, the horizontal transfer matrix from m1 to m4 is

$$M_x = B_4 R(\mu_x) B_1^{-1}, \qquad B = \begin{pmatrix} \sqrt{\beta} & 0 \\ -\alpha/\sqrt{\beta} & 1/\sqrt{\beta} \end{pmatrix}, \tag{6}$$

with $R(\mu_x)$ a rotation by the horizontal phase advance and $B_{1,4}$ the boundary Courant–Snyder parameters of Table 1. Let J be the covariance of the non-dispersive part of the shot centroid at m1 (position 0.5 mm and angle 0.2 mrad rms, uncorrelated); the momentum jitter $\sigma_{\delta,j}$ is added separately through the dispersion. The centroid therefore reaches the exit with covariance $M_x J M_x^{\mathsf{T}}$ plus the momentum term $\sigma_{\delta,j}^2 \mathbf{d}_4 \mathbf{d}_4^{\mathsf{T}}$, which is the same for every matched setting. In this nominal model the quadrupoles influence the exit jitter only through μ_x. Over a full period the exit position spread ranges from 0.26 mm to 0.95 mm and the angle spread from 0.12 mrad to 0.41 mrad, in opposite phase (Fig. 4): a matched line can move booster jitter between position, which the septum blade sees, and angle, which the injection kicker sees, but cannot remove it. On error realizations, the residual trajectory after steering, the optics errors and roll coupling add to this jitter; the test losses are computed with the full transport of each realization (Sections 4.2 and 4.3). Averaged over the Gaussian jitter in the linearized model, these static contributions raise the blade loss of the error-free line by only a few percent (Section 4.5); the different tracked losses of some test populations reflect the jitter tail sampled by their 600 shots.

3.3. Loss-based observable and design

For a Gaussian projection with centroid $\bar{u}$ and rms size σ at an aperture with edges $a_\pm$, the lost fraction is

$$P = \Phi\left(\frac{a_- - \bar{u}}{\sigma}\right) + \Phi\left(\frac{\bar{u} - a_+}{\sigma}\right), \tag{7}$$

with Φ the standard normal distribution function. Equation (7) is exact in each plane for the rectangular and one-sided apertures, including the septum blade. For the round apertures of the pipe and quadrupoles we evaluate instead the exact probability that the Gaussian (x, y) distribution lies outside the circle, using a closed-form radial integral and periodic angular quadrature. Two projected tails would underestimate it: for a centered isotropic beam at a circle of radius 5σ the exact loss is $e^{-12.5} = 3.73 \times 10^{-6}$, whereas $4\Phi(-5) = 1.15 \times 10^{-6}$. Within the linearized Gaussian model, the line-loss probability for a shot is bounded below by the largest of these event probabilities and above by their sum. We use the lower bound as the prediction and design surrogate. These bounds apply to model probabilities; finite particle counts fluctuate, and nonlinear tracking can also differ from the linearized prediction. The blade at the injection-septum entrance and at the line end gives the largest event in every evaluated case, and every tracked first loss occurs in the injection septum. When the centroid is itself Gaussian, as for linearly transported jitter, the shot-averaged lost fraction is Eq. (7) with the summed covariance of beam and jitter. This makes a smooth, loss-relevant objective. The loss-based setting minimizes the logarithm of the largest shot-averaged P on the error-free line, subject to the six exit equalities, horizontal peak beta ≤ 60 m and the strength bounds, using sequential least-squares programming (SLSQP) [17] with a 200-iteration budget and termination tolerance 10^{-10}. Three initial settings are evaluated in the bounded search. The objective uses the nominal entrance beta functions and the error-free line; the static residuals and the 5 % beta-function fluctuations of the test are not part of it, so the error-free prediction and the tracked test describe different populations. The setting is frozen before a fresh test on 200 steered error realizations, evaluated with tracked

particles. Paired differences between settings are averaged per error realization; the pointwise 95 % intervals are percentile intervals of 10 000 bootstrap resamples of the 200 realizations. This realization-cluster bootstrap accounts for the dependence between the shots of one realization, but it does not certify that the heavy loss tail is converged; it describes the sampled population of realizations and shots.

Four loss quantities appear below, and they are conditioned differently. The design objective $\bar{P}$ is the shot-averaged loss of the error-free line with the nominal beam. The jitter-integrated loss applies the same Gaussian average over the entrance jitter to each steered realization, within the linearized transport and with the nominal beam covariance, and uses the largest single-aperture probability as its event. The conditional prediction evaluates Eq. (7) for each sampled shot, with its actual centroid and beta factors, and averages over the shots. The observed loss is the tracked particle count. Only the last two describe the same shots. We report the gain of one setting over another as the ratio of their mean jitter-integrated losses over the realizations of a population, with a percentile interval from 10 000 paired bootstrap resamples that draw the same realization indices for both settings. The primary population is the second fresh test of Section 4.3; it was fixed before that test was run with shared reading errors.

3.4. Accuracy and spatial observation resolution

Independent hard-edge maps reproduce AT tracking at every element boundary within 1.6×10^{-8}, against a tolerance of 2×10^{-6} in the stated SI convention. On the checked error cases, centered-map, integration-step, probe-step and quadrature checks change clearances by at most 0.6858 µm, far below the reported effects, and the sampled peak beta functions change by less than 0.35 mm between 32 and 64 subdivisions per element. These are finite-case checks, not universal error bounds or loss-tail certificates; their details, including the audit of the earlier four-station campaign, are given in the supplement. In the low-beta setting the second quadrupole of the third triplet, `q32`, remains at its lower bound $-3\,\mathrm{m}^{-2}$.

4. Results

4.1. Active restrictions of the element-wise line

Figures 2 and 3 compare the baseline setting with the optimized low-beta setting. The baseline strengths are the quadrupole settings supplied with the BTS configuration before this study; no record states the exit target for which they were designed, and they do not satisfy the present one (Table 3). They are therefore a starting point, not a competing design. Their nominal five-sigma envelope extends across the injection-septum blade boundary, but this is expected: the blade distance is defined from the matched exit beam (Section 2.3), so any setting with a much larger exit beta function overlaps it. The optimized setting satisfies the exit target, reduces the large vertical envelope, and restores positive nominal clearance. Table 3 gives the corresponding exit lattice functions, peak beta values and minimum clearances. The figure comparison therefore includes the improvement from matching the baseline BTS line, rather than holding the exit optics fixed for both curves.

After matching, a separate comparison of the matched reference and optimized low-beta setting isolates the effect of interior beta reduction. In that matched comparison the blade clearances are 0.648 mm at the injection-septum entrance and 0.683 mm at the line end (m4), essentially equal for both settings. The nearest interior restriction is the vertical gap of b2, with 9.12 mm for the matched reference and 9.32 mm for the optimized low-beta setting. The bend-gap and extraction-septum aperture sweeps leave the blade as the limiting restriction for these matched settings. Matching is required for nominal delivery, and further reductions of interior beta at a fixed exit target act mainly on clearance away from the blade. The error-study tables below retain this matched reference as a separate comparison setting; they do not report an error ensemble for the baseline setting.

Table 3: Nominal optimization of the baseline BTS setting. Peak beta functions are sampled with the same spatial resolution as Figure 2. The maximum normalized exit residual is evaluated using the declared optical scales; a matched setting requires it to be below 10^{-3}. A negative clearance means that the projected five-sigma envelope overlaps an aperture boundary, not that every particle is lost.

Observable	Baseline setting	Optimized low beta
Exit β_x [m]	44.9808	2.3365
Exit β_y [m]	242.6069	4.2562
Peak β_x [m]	52.2506	33.7945
Peak β_y [m]	242.6069	28.4371
Minimum clearance [mm]	-1.5170	0.6484
Max. normalized exit residual	4.77×10^{3}	2.88×10^{-10}

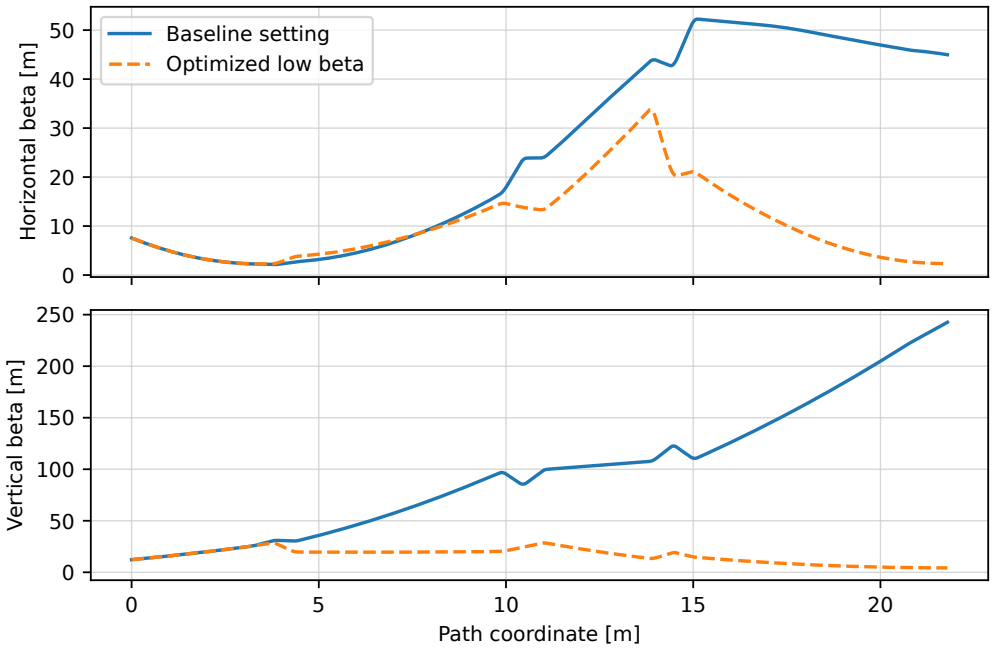


Figure 2: Baseline setting and optimized low-beta lattice functions, drawn with Accelerator Toolbox linear-optics plotting from the same entrance Courant–Snyder parameters and with the same observation resolution. The baseline setting uses the unmodified configuration strengths and does not satisfy the exit target; the optimized setting is matched. Both curves use the same magnet positions and bend geometry.

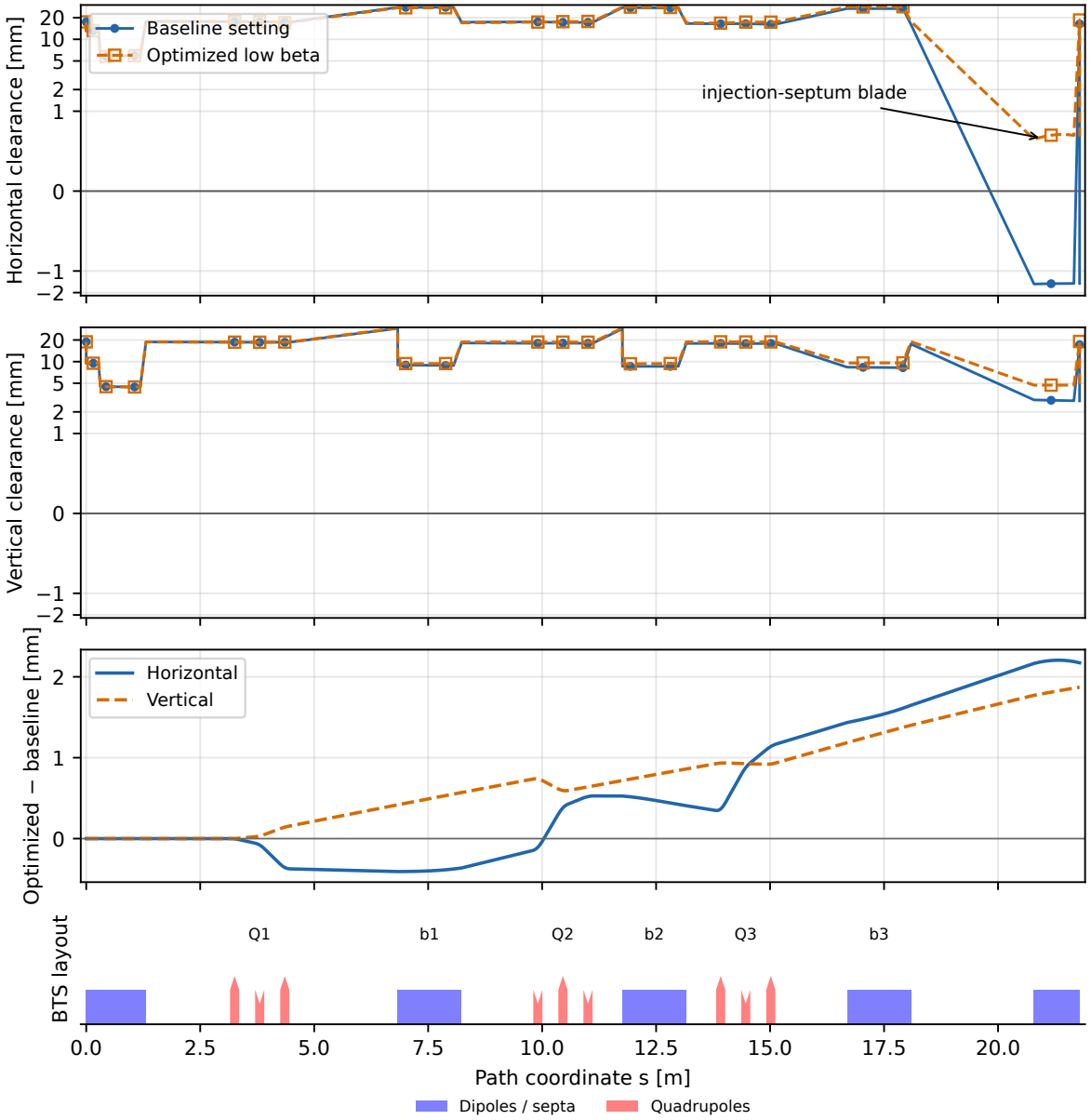


Figure 3: Nominal five-sigma clearance for the baseline setting (solid blue with filled circles) and optimized low beta (dashed orange with open squares). A symmetric logarithmic scale, linear near zero, displays the baseline setting's negative horizontal clearance at the injection septum. Zero marks the five-sigma envelope touching an aperture. The third panel shows optimized minus baseline clearance in each plane. The bottom panel gives the common BTS layout; Q1–Q3 identify quadrupole triplets. The baseline and optimized settings share the entrance conditions but have different exit lattice functions. Design beam, no errors.

4.2. Trajectory correction

The earlier steering comparison (supplement Section 5) evaluates the uncorrected and corrected matched lines on 200 error realizations with three shots of 2 000 particles each; both settings and both variants see the same magnet errors, shots and particles. Without correction the static errors displace the exit centroid by 1.2 mm rms, and 10.08 % of the particles are lost; every recorded first loss is at the injection-septum blade. With the dipole trims the exit position spread falls to 684 µm, the angle spread stays at 367 µrad (364 µrad uncorrected), and the loss falls to 1.68 % (1.47 % for the low-beta setting); the rms trim is 204 µrad. Two trims close to the exit, b3 and the injection septum, steer both position and angle at the septum. The fit uses position readings only, so the exit angle is constrained only through them. The uncorrected vertical exit centroid is 1116 µm and 100 µrad rms for the matched reference (Table 4); it stays inside the 5 mm vertical restrictions but is the static vertical injection error passed to the injection calculation.

Two observations arise from these results. First, after correction the exit centroid spread, about 0.68 mm and 0.37 mrad, is mainly that of the booster jitter (0.64 mm and 0.33 mrad for the error-free matched reference, Table 7); averaged over the Gaussian jitter in the linearized model, the static residual raises the loss by only a few percent (Section 4.5). Second, the mean minimum clearance does not rank the variants by loss: the uncorrected line has the largest mean clearance (0.584 mm against 0.524 mm) and six times the loss. All 600 shots enter both means, and the moments are uncensored by apertures, so this is not a survivor effect. Without correction the static errors leave a mean exit offset of −68.5 µm, away from the blade (8.8 µm after correction), which raises the mean clearance, while the spread of the exit centroid grows from 0.68 mm to 1.20 mm. At a one-sided edge a with fixed beam size σ and centroid spread τ, the mean clearance $a - E[\bar{u}] - 5\sigma$ does not depend on τ, whereas the mean loss, approximately $\Phi\big(-(a - E[\bar{u}])/\sqrt{\sigma^2 + \tau^2}\big)$, grows with it; the same mean clearance can therefore hide very different tail losses. The mean Gaussian loss prediction, by contrast, agrees with the tracked loss fraction of each row within 0.4 % relative. Because the tracked beam is itself Gaussian, this agreement checks the moment transport and loss accounting; it does not test non-Gaussian booster tails, which would raise the blade loss most where the blade is many beam sizes away. In this earlier steering comparison, the low-beta row loses 0.21 percentage points less than the matched reference (paired interval [−0.39, −0.06] percentage points). Reading errors were drawn separately for each setting, so this contrast does not isolate the low-beta effect. Its mean clearance change, −0.001 mm ([−0.009, 0.007] mm), is unresolved, as is its loss change in the earlier fresh test retained in supplement Section 5. The controlled, shared-reading primary test in Section 4.3 gives a resolved change of −0.13 [−0.24, −0.04] percentage points. Its horizontal phase advance is about 1° smaller than that of the matched reference, which lowers its error-free blade loss from 1.28 % to 1.16 % (Table 5); in the nominal model the lower vertical peak beta does not enter the horizontal blade loss.

Table 4 repeats the steering on the same realizations with shared readings, with fewer readings, and with b1–b3 trimmed as one series-powered knob whose kick in each bend is proportional to its bending angle. The matched-reference rows of the full layout reproduce the earlier steering results exactly. The loss-based setting loses 0.38 % here, against 0.23 % with its own reading errors in the earlier comparison, so the reading errors alone move its tracked loss by more than half. Reading only the exits of the three triplets, the septum entrance and the line end changes the losses by less than 0.01 percentage point. Series-powered bends leave one knob fewer near the exit, and the exit angle spread grows from 0.37 mrad to 0.44 mrad for the matched reference and from 0.41 mrad to 0.50 mrad for the loss-based setting. Raising the weight of the septum-entrance and line-end readings from 3 to 10, to constrain the exit angle more strongly, changes the exit angle of the loss-based setting from 0.413 mrad to 0.411 mrad and the losses by less than 0.01 percentage point. With the assumed 50 µm reading errors, this reweighting gives little improvement in the exit-angle rms. Other estimators, monitor layouts and direct angle measurements were not tested.

Table 4: Steering layouts on the 200 realizations of the earlier steering study (supplement Section 5), with reading errors shared by both settings (tracked, 3 shots of 2 000 particles). Readings: quadrupole exits (all nine or the three triplet exits), septum entrance and line end; in the last layout the last two are weighted 10 instead of 3. Knobs: six independent dipole trims, or four with b1–b3 in series. Integr.: jitter-integrated loss. Exit centroids are rms over shots; the vertical plane is not corrected.

Readings / knobs	Setting	Lost [%]	Integr. [%]	x_{rms} [mm]	x'_{rms} [mrad]	y_{rms} [mm]	y'_{rms} [mrad]
11 / 6	Matched	1.68	1.32	0.68	0.37	1.12	0.100
	Loss-based	0.38	0.29	0.54	0.41	0.77	0.094
5 / 6	Matched	1.67	1.32	0.68	0.37	1.12	0.100
	Loss-based	0.38	0.29	0.54	0.41	0.77	0.094
11 / 4	Matched	1.65	1.42	0.69	0.44	1.12	0.100
	Loss-based	0.39	0.35	0.55	0.50	0.77	0.095
5 / 4	Matched	1.58	1.39	0.69	0.44	1.12	0.100
	Loss-based	0.40	0.35	0.55	0.49	0.77	0.095
11 / 6, ends ×10	Matched	1.67	1.32	0.68	0.37	1.12	0.100
	Loss-based	0.38	0.29	0.54	0.41	0.77	0.094

4.3. Phase advance of the matched settings and the loss-based setting

Bounded least-squares matching from 360 random starts finds 349 matched quadrupole settings within the strength bounds, with horizontal phase advances between 200.2° and 232.0°. To test whether this range reflects the search or the matching problem, we scanned μ_x in 2° steps as a seventh matching condition, and searched for the smallest and largest μ_x subject to the six exit equalities, the strength bounds and peak beta functions of at most 60 m in both planes, from five starts per edge (Fig. 5). Matched settings are found on the grid from 202° to 232°. The lower edge, 200.2°, is reached by three converged searches with no strength bound and neither peak-beta limit active. A small study that changes only the exit dispersion target moves it. With D_x fixed, lowering D'_x at m4 from 47.5 mrad to 32, 16 and 0 mrad lowers the edge to 194.9°, 189.5° and 184.1°, again with no bound or peak limit active. At 32 mrad and 16 mrad, all three starts give matched candidates near the reported phases. At $D'_x = 0$, two starts give matched candidates; the start from the dispersion-free scan setting fails the declared matching test. The matched searches at these changed D'_x targets all stop at their iteration limit, so these phases are best-found values. Setting $D_x = 0$ at the configured D'_x raises the edge to 203.2°, where the vertical 60 m limit is active. The shift of the best-found lower phase with D'_x indicates that the inherited exit dispersion conditions restrict the phase range explored by these local searches. It does not identify a unique binding cause or certify an edge globally. The upper edge, 232.0°, is set by the 60 m limit in both planes. These are ranges found by sampled and local searches, not a proof of global reachability. The matched reference (211.8°) and low-beta (210.8°) settings lie near the middle. As Eq. (6) implies, the predicted blade loss of the error-free line depends on the setting through μ_x alone: it falls monotonically from 4 % at the upper edge to 0.27 % at the lower edge, while the peak beta functions vary independently between about 23 m and 60 m. Reducing the loss therefore means choosing the smallest matchable μ_x.

The loss-based search of the earlier study ended at that edge, $\mu_x = 200.2°$, although all three of its SLSQP starts reached the 200-iteration limit and only one was feasible. The converged edge solution has the same phase advance to within 8×10^{-7}° and the same peak beta to within 0.02 m; we therefore regard the loss-based setting as the best-found feasible setting at the lower edge of the matched range, not as a certified global optimum. Its predicted shot-averaged loss is 0.27 % against 1.28 % for the matched reference, because its exit position jitter is 506 µm instead of 643 µm.

An earlier fresh test of the matched reference, low-beta and loss-based settings, on 200 other realizations with reading errors drawn per setting, is given in the supplement. Its tracked gain was 3.3, the low-beta change was unresolved, and the mean minimum clearance did not resolve the loss reduction. The price of the loss-based setting is an exit angle jitter larger by 13 % (368 µrad

against 327 µrad), which the injection kicker must accept.

Table 5 gives a second test on 200 independently seeded realizations, with the reading errors shared by all settings. Of 1 200 000 tracked particles the matched reference loses 11673, low beta 10104 and the loss-based setting 942; the loss-based change is −0.89 [−1.50, −0.41] percentage points. Here the loss-based setting is the converged edge solution. The earlier loss-based setting has the same nominal horizontal transfer to the reported precision and, with shared readings, loses exactly the same particles, as does a second copy of the same configuration. When each setting had its own reading errors, the same realizations gave 1 022 and 1 564 lost particles for these two settings, so that difference came from the reading errors. Low beta loses less than the matched reference on this population, −0.13 [−0.24, −0.04] percentage points.

On this primary population, the mean jitter-integrated loss is 1.32 % for the matched reference and 0.295 % for the loss-based setting, a gain of 4.48 (95 % interval 4.33–4.63). The interval resamples the error realizations only; the Gaussian jitter is integrated at the assumed level and the nominal beam covariance is fixed. It excludes uncertainty in the model and input assumptions. On the realizations of Table 4 the same estimator gives 4.59 (4.46–4.72). The tracked losses of the earlier test populations, and their gains of 3.3 to more than 10, come from 600 sampled shots each and vary mainly with the sampled jitter tail. The tracked counts test the shot-conditional prediction. For the matched reference and low beta they lie within half a Poisson standard deviation of it (11673 against 11705, 10104 against 10154). The loss-based setting loses 942 particles against 880 expected: the prediction is 6.5 % below the observed count. The difference is 2.1 standard deviations under the diagnostic Poisson approximation $(N_{\mathrm{obs}} - N_{\mathrm{exp}})/\sqrt{N_{\mathrm{exp}}}$, which excludes model-systematic uncertainty. All 942 observed losses occur in 16 of the 600 sampled shots. One near-blade shot loses 454 particles against 410.9 expected, accounting for about 70 % of the aggregate excess. A nonlinear-moment check of this shot was not performed, so the cause remains unresolved. The count diagnostic does not establish a general accuracy tolerance.

Table 5: Second fresh test: 200 steered error realizations with new seeds and shared reading errors, 3 shots of 2 000 particles each. $\bar{P}$: predicted loss of the error-free line; Integr.: jitter-integrated loss on the steered realizations; expected lost particles: Gaussian prediction conditional on the sampled shots; observed: tracked count; last column: paired difference of the tracked loss from the matched reference with its pointwise 95 % realization-cluster interval. Loss-based is the converged phase-edge setting; the earlier loss-based setting and a duplicate configuration give identical rows. Disp.-free is the best dispersion-free scan setting.

Setting	μ_x [deg]	$\hat{\beta}$ [m]	$\sigma_{x',j}$ [mrad]	$\bar{P}$ [%]	Integr. [%]	Lost particles expected	observed	Lost − Matched [pp]
Matched	211.8	51.3	0.327	1.28	1.32	11705.2	11673	—
Low beta	210.8	33.8	0.331	1.16	1.21	10153.7	10104	−0.13 [−0.24, −0.04]
Loss-based	200.2	48.1	0.368	0.27	0.30	880.5	942	−0.89 [−1.50, −0.41]
Disp.-free	188.0	35.6	0.393	4.7×10^{-3}	6.3×10^{-3}	0.5	1	−0.97 [−1.64, −0.43]

4.4. Exit target and dispersion

At m4 the configured target gives the injected beam an rms divergence of 76 µrad: 57 µrad from the betatron emittance and 50 µrad from $D'_x\sigma_\delta$. The 0.1 % shot energy jitter adds a centroid angle of 47 µrad and a position of 81 µm rms. Both terms are fixed by the target, the same for every matched setting, and absent for a dispersion-free exit.

The configured exit dispersion is that of the injected beam in the configuration of the NKM injection study; we have found no derivation of it from the ring optics, whose dispersion at the injection kicker is nearly zero. In the local study of Section 4.3, reducing the prescribed slope extends the matched settings found toward smaller phase advances, where less jitter reaches the blade. Matching the same Courant–Snyder parameters with $D_x = D'_x = 0$ at m4 is feasible within the strength bounds: all 115 solutions from 120 bounded random starts keep both peak betas below 60 m (smallest larger peak 26.3 m). The constrained searches place the dispersion-free range between 187.2° and 214.5° (Fig. 5); its upper edge is set by q22 at its strength bound. At

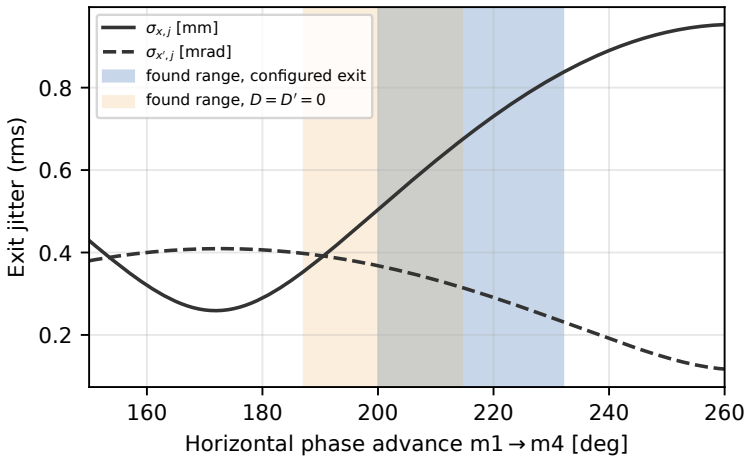


Figure 4: Rms exit position (solid, mm) and angle (dashed, mrad) jitter from Eq. (6) versus horizontal phase advance for the assumed booster jitter, with the phase-advance ranges found by the constrained searches for the configured exit (blue) and a dispersion-free exit (orange; its momentum term differs slightly). The blade loss of the matched settings is shown in Fig. 5.

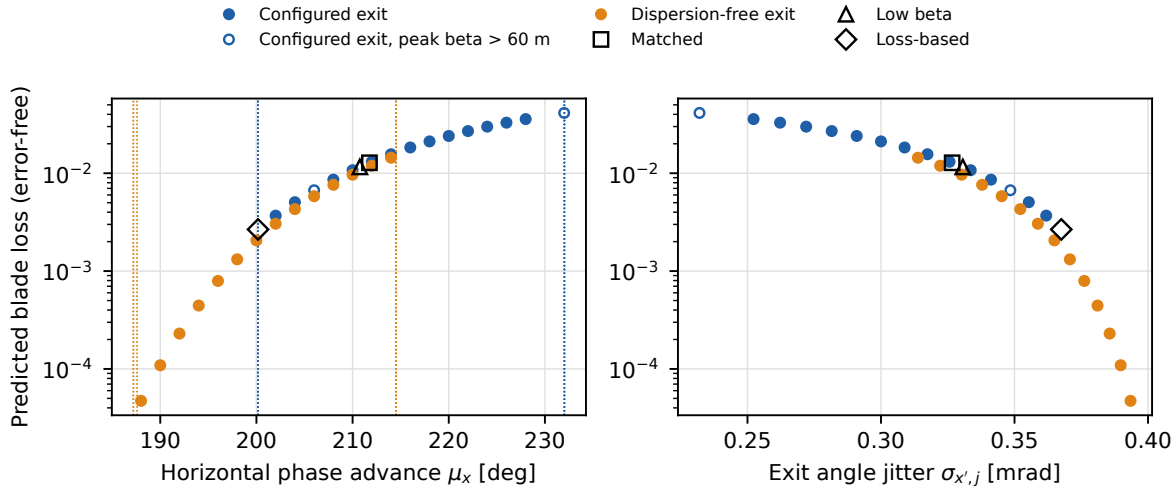


Figure 5: Phase-advance scan. Left: predicted blade loss of the error-free line for the lowest-peak-beta matched setting at each scanned μ_x, for the configured exit (blue) and a dispersion-free exit (orange); open symbols exceed the 60 m peak-beta limit. Dotted lines mark the edges found by constrained searches. Right: the same settings against the exit angle jitter. Black symbols mark the matched reference, low-beta and loss-based settings.

the lower edge the searches that reach 187.2° stopped at their iteration limit with no constraint active, and the one converged search ends at 187.6° on the vertical 60 m limit. The range includes phases where booster jitter falls mainly on the exit angle. The best scanned setting ($\mu_x = 188°$, larger peak beta 35.6 m) has a predicted error-free blade loss of 4.7×10^{-3} % and an exit angle jitter of 393 µrad; the beam divergence falls to 57 µrad. On the second test it loses 1 of 1 200 000 tracked particles, a fraction of 8.3×10^{-7}, where the prediction conditional on the sampled shots expects 0.51 particles. Its jitter-integrated loss corresponds to 76 particles; most of that integral comes from large-jitter shots that the 600 sampled shots do not contain, so this test cannot check the integrated tail at a stated precision. Non-Gaussian booster tails, if present, could raise these small losses; they are not modeled.

Among the alternatives tested here, the exit target therefore has the largest leverage on injection. Removing the exit dispersion lowers the injected beam's divergence by a quarter and lets the line put booster jitter where the septum blade does not see it, but the angle jitter, 393 µrad against 327 µrad for the matched reference, then arrives at the injection kicker. In the NKM injection study [5], the kicker's 80 % angular acceptance is 0.44 mrad, and once the static vertical injection error left by transfer-line magnet errors is removed, booster position and angle jitter are the dominant kicker-specific capture losses. This line has no vertical correction; correcting that vertical error will be addressed in a future study. Whether the trade favors the blade or the kicker must be decided with the ring, not within the line.

4.5. Sensitivity to the error levels

The 0.5 mrad rms quadrupole roll is large for a transfer line, and 0.1 % rms shot energy jitter is at the high end of booster performance. On another 200 independently seeded realizations, rolls of 0.1 mrad to 0.5 mrad change the predicted loss of every setting by less than 0.001 percentage point, and halving the energy jitter changes it by at most 0.04 percentage point; the supplement gives the table. Rolls couple a small part of the horizontal motion into the vertical plane, whose restrictions keep several millimeters of clearance, and the 81 µm exit contribution of the energy jitter is small compared with the betatron jitter.

The jitter and blade-distance sweeps below retain the earlier loss-based setting and reading errors drawn separately for each setting. They are sensitivity studies; the primary comparison in Table 5 uses the converged edge setting and shared reading errors. The residual table uses the matched reference from the same earlier jitter sweep.

For each steered realization we also evaluate the jitter-integrated loss: Eq. (7) averaged over the Gaussian entrance jitter on that realization's steered line, within the linearized transport and the nominal beam covariance. It resolves losses far below the reach of sampled shots. Table 6 scales the position and angle jitter together on a further 200 independently seeded realizations. At the assumed level the jitter-integrated losses are 1.31 % and 0.28 % for the matched reference and loss-based settings, close to the error-free 1.28 % and 0.27 %: the steered static residual adds only 2 % and 5 %, and the loss-based gain is 4.7. Halving the jitter lowers the losses by more than two orders of magnitude, with a gain of 31. At 0.1–0.2 of the assumed level, i.e. 13–44 % of the beam size at m1, every setting loses a fraction of at most 3×10^{-11}, and the loss-based setting is no better than the others; for it, the static residual then contributes more than the jitter. The loss reduction reported above is therefore a property of large booster jitter.

Table 7 compares the rms exit centroid of the steered realizations with the nominal jitter alone. The scale applies to the betatron jitter; the 0.1 % momentum jitter stays, so the scale-0 row contains the static residual and the momentum jitter. At the assumed jitter the two columns agree, as Section 3.2 predicts. At 0.1 of it the horizontal exit angle is 0.18 mrad rms, whereas the nominal jitter gives only 0.057 mrad: the angle left by steering is three times the jitter part and hardly changes between scales 0 and 0.2. The vertical centroid remains at 1.07 mm and 0.10 mrad rms in this sweep. The vertical trajectory is uncorrected; among the three settings it ranges from 0.78 mm to 1.07 mm at scale 0. At low jitter, then, the phase advance still divides

the jitter between position and angle, but the delivered centroid is dominated by the horizontal steering residual and the uncorrected vertical trajectory.

Table 6: Dependence on the booster betatron jitter. Position and angle jitter at m1 are scaled together from the assumed 0.5 mm and 0.2 mrad (scale 1) and are given in units of the rms beam size and divergence at m1; the momentum jitter stays at 0.1 %. The earlier loss-based setting and per-setting reading errors are retained. Losses are the jitter-integrated predictions averaged over 200 steered error realizations, and the last column is the error-free line (analytic, not tracked).

Jitter	At m1 [beam sizes]		Jitter-integrated loss, steered [%]			
scale	position	angle	Matched	Low beta	Loss-based	Error-free matched
0	0.00	0.00	2.7×10^{-13}	3.8×10^{-13}	2.5×10^{-10}	3.4×10^{-14}
0.05	0.07	0.11	4.4×10^{-13}	5.7×10^{-13}	2.8×10^{-10}	4.9×10^{-14}
0.1	0.13	0.22	2.8×10^{-12}	2.7×10^{-12}	3.8×10^{-10}	3.0×10^{-13}
0.2	0.26	0.44	2.6×10^{-9}	1.9×10^{-9}	1.3×10^{-9}	7.9×10^{-10}
0.5	0.66	1.10	3.2×10^{-3}	2.3×10^{-3}	1.0×10^{-4}	2.6×10^{-3}
1	1.32	2.21	1.3×10^{0}	1.2×10^{0}	2.8×10^{-1}	1.3×10^{0}

Table 7: Rms exit centroid of the matched reference on the 200 steered realizations of Table 6 against the nominal exit jitter of the error-free line, by betatron jitter scale, from the earlier sweep with reading errors drawn separately for each setting. The 0.1 % momentum jitter is kept at every scale. Analytic centroids of three sampled shots per realization.

Jitter	Steered realizations (rms)				Nominal jitter only	
scale	x [mm]	x' [mrad]	y [mm]	y' [mrad]	$\sigma_{x,j}$ [mm]	$\sigma_{x',j}$ [mrad]
0	0.096	0.176	1.073	0.096	0.081	0.047
0.1	0.113	0.179	1.073	0.096	0.103	0.057
0.2	0.156	0.187	1.073	0.096	0.151	0.080
0.5	0.328	0.238	1.073	0.096	0.329	0.168
1	0.639	0.366	1.073	0.096	0.643	0.327

The centroid-to-blade distance $d_b = 1.478$ mm is taken from the NKM injection geometry, and the blade loss depends on it roughly exponentially. Figure 6 repeats the jitter-integrated evaluation on the same 200 steered realizations for d_b from 1 mm to 3 mm. At the assumed jitter, a loss below 0.1 % requires $d_b \geq 2.05$ mm for the matched reference, 2.01 mm for low beta and 1.64 mm for the loss-based setting; at $d_b = 1$ mm the losses are 6.6 % and 3.0 %. With a tenfold smaller jitter the loss stays below 2×10^{-6} even at 1 mm. A larger d_b places the injected beam farther from the stored beam and so demands more of the ring's dynamic acceptance at injection, which must be evaluated with the injection calculation [5]. A thinner blade, such as the 1 mm septum of the TDR design [2], would either add 1 mm to d_b at the same injection offset or allow the same d_b with the injected beam 1 mm closer to the stored beam.

5. Discussion and conclusions

The baseline comparison illustrates the required boundary match. The blade distance is defined from the matched beam, and the unmatched baseline overlaps it. The design comparison concerns settings that already share the prescribed endpoint optics.

In a BTS line matched at both ends, the nominal beam size at the ends is fixed and so is the nominal clearance of every restriction outside the quadrupole section. With the blade geometry of the NKM injection study, the injection septum is that restriction, and it limits the line by an order of magnitude more than any interior aperture. Lowering the interior peak beta, the usual design heuristic, therefore changes only clearances of several millimeters. In the controlled shared-reading primary test, low beta changes the loss by -0.13 [-0.24, -0.04] percentage points relative to the

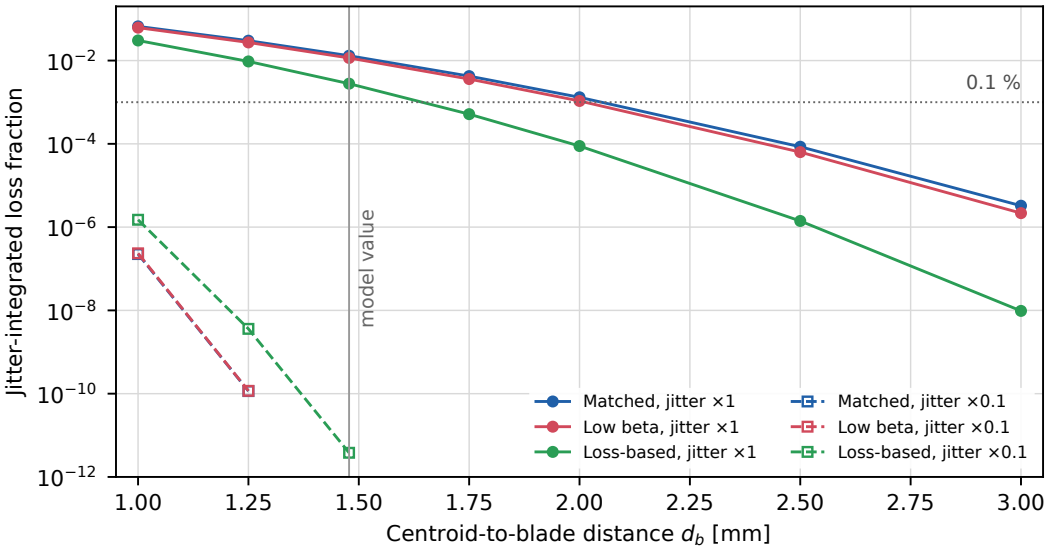


Figure 6: Jitter-integrated blade loss against the centroid-to-blade distance d_b for the three matched settings, including the earlier loss-based setting, with per-setting reading errors, at the assumed booster jitter (solid, filled) and a tenfold smaller jitter (dashed, open), averaged over 200 steered error realizations (analytic). Values below 10^{-12} are omitted. The vertical line marks the model value 1.478 mm and the dotted line a 0.1 % loss.

matched reference. The earlier steering comparison gives a decrease of 0.21 percentage points with per-setting reading errors; it does not isolate the optics effect. The controlled benefit accompanies a horizontal phase advance about 1° smaller, rather than a direct dependence on the vertical peak beta.

The quadrupoles act on the loss mainly through the trajectory. Static trajectory errors are reduced by field trims of the existing dipoles, without adding any element to the line; the two trims nearest the exit steer both position and angle at the septum. A horizontal residual, the optics errors and the uncorrected vertical trajectory remain. At the assumed stress level, the dominant remaining contribution is booster shot jitter, and in the nominal matched line its arrival at the exit depends only on the horizontal phase advance. The line chooses whether the jitter appears as position at the blade or as angle at the injection kicker, but it cannot remove it. For the configured exit, the matched settings found span about 32° of phase advance; the error-free blade loss depends on μ_x alone and is smallest at the lower edge found for that range. The local dispersion study shows that the best-found lower phase decreases with the exit dispersion slope inherited from the companion configuration; the iteration-limited searches do not establish a global edge. On the primary test population the best-found setting there has a mean jitter-integrated loss 4.48 times smaller than the matched reference (4.33–4.63), at 13 % more angle jitter. A dispersion-free exit extends the range to smaller μ_x and lowers the loss much further, again at the cost of angle jitter. The choice between blade loss and injection-angle jitter belongs to the injection design as a whole: for the NKM, whose 80 % angular acceptance in the companion study is 0.44 mrad, the exit-angle jitter of 0.33 mrad–0.39 mrad at the assumed booster jitter is of the same order, so at that level the trade must be evaluated with ring capture.

Two methodological points carry over. The mean minimum five-sigma clearance does not rank settings by loss: it rewards static offsets away from a one-sided restriction and is blind to the jitter tail. A Gaussian loss probability from the propagated centroid and covariance is smooth and can be optimized directly. Conditioned on the sampled shots, it matches the tracked counts within 1 % relative in the steering and first test populations and within half a Poisson standard deviation for the matched reference and low beta in the second test. For the loss-based setting, the prediction is 6.5 % below the observed count, with the discrepancy still unresolved. The phase-advance view also gives a quick test for any matched transfer line: compute μ_x, read

off where the booster jitter lands, and compare with the restriction and the kicker acceptance.

The booster jitter assumed here is 1.3 and 2.2 beam sizes at m1, and the blade loss and the benefit of the loss-based setting depend on it steeply. At that stress level the phase advance is the design knob for blade loss. At 0.1–0.2 of the level every matched setting loses less than 10^{-10}, and the delivered beam error is set by the horizontal angle left by steering, about 0.18 mrad, and by the uncorrected vertical trajectory, about 1.07 mm for the matched reference. For the tested settings and correction scheme, changing the matched optics leaves these residuals as the main delivery errors. The vertical trajectory is uncorrected in every setting because the modelled line has no vertical corrector. In this low-jitter regime, correction and instrumentation, especially in the vertical plane, therefore matter more for this line than the choice among the tested matched optics. Finite-shot losses are heavy-tailed, and their population means vary with the sample. The single-loss dispersion-free test does not certify its integrated tail. Aperture checks at element entrances and the line end bracket the septum channel without resolving continuous interception. The study does not certify real transmission or derive construction tolerances, ring capture or kicker transparency.

Statements and Declarations

Funding.. National Research Foundation of Korea (NRF) grant funded by the Korean government (MSIT), NRF Grant No. RS-2022-00143178.

Competing interests.. The author declares no competing interests.

Author contribution.. Chong Shik Park is responsible for scientific interpretation and final article approval; final approval is pending.

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